

Product Dataset – Data Cleaning & Transformation Report

Project Objective:

A cleaned and transformed product dataset where missing values, duplicates, inconsistent formats, and text issues were identified and corrected using Excel formulas and functions.

Dataset Description:

A structured Excel dataset containing product information such as Product ID, Product Name, Brand Name, Price, Quantity, and Category. This project demonstrates **data cleaning, transformation, and preprocessing techniques** performed on the dataset to prepare it for analysis.

Column Description:

Column Name	Description
Product ID	ID Numbers
Product Name	Name of the Product
Brand Name	Brand name of Product
Price	Price of the Product
Quantity	No of Quantity by Product
Category	Category based on the Product

Data Cleaning Steps Performed:

1. Handling Missing Values

- ✓ Checked for missing values in **Price** and **Category** columns.
- ✓ **Result:** No missing values found.

2. Splitting & Merging Data

- ✓ Split *Product ID* into: Manufacturing Date & Country code

Formulas Used:

=LEFT(A2, 6)

=RIGHT(A2, 2)

=TEXTJOIN(" ", TRUE, B2, C2)

3. Correcting Inconsistent Data

- ✓ Checked for inconsistent text formats in **Product Name**.

4. Removing Duplicates

- ✓ Identified duplicate rows using conditional formatting.
- ✓ **Result:** There are 3 duplicate data found in product ID column.

5. Number Formatting

- ✓ Converted **Price** column to currency format.
- ✓ Converted **Manufacturing Date** to proper date format.

6. Conditional Formatting

- ✓ Applied color scales to Price column.
- ✓ Highlighted **Electronics** category using a custom rule.

7. Standardizing Text Formats

- ✓ Used Find & Replace to fix typos or inconsistent text.
- ✓ **Result:** No typos found in Product Name or Category.

Conclusion:

The data-cleaning project successfully transformed a raw product dataset into a structured, accurate, and analysis-ready format. Through a series of systematic steps—including handling missing values, correcting inconsistent text formats, splitting and merging fields, removing duplicates, applying number formatting, and using conditional formatting—the dataset was fully standardized and validated.

The final output now includes enhanced fields providing deeper insights and improved usability for reporting or analytics. With all inconsistencies resolved and duplicates removed, the dataset is now clean, reliable, and ready for further analysis, visualization, or integration into business intelligence tools.

This project demonstrates strong proficiency in Excel-based data preprocessing and showcases the importance of structured cleaning workflows in maintaining high-quality datasets.